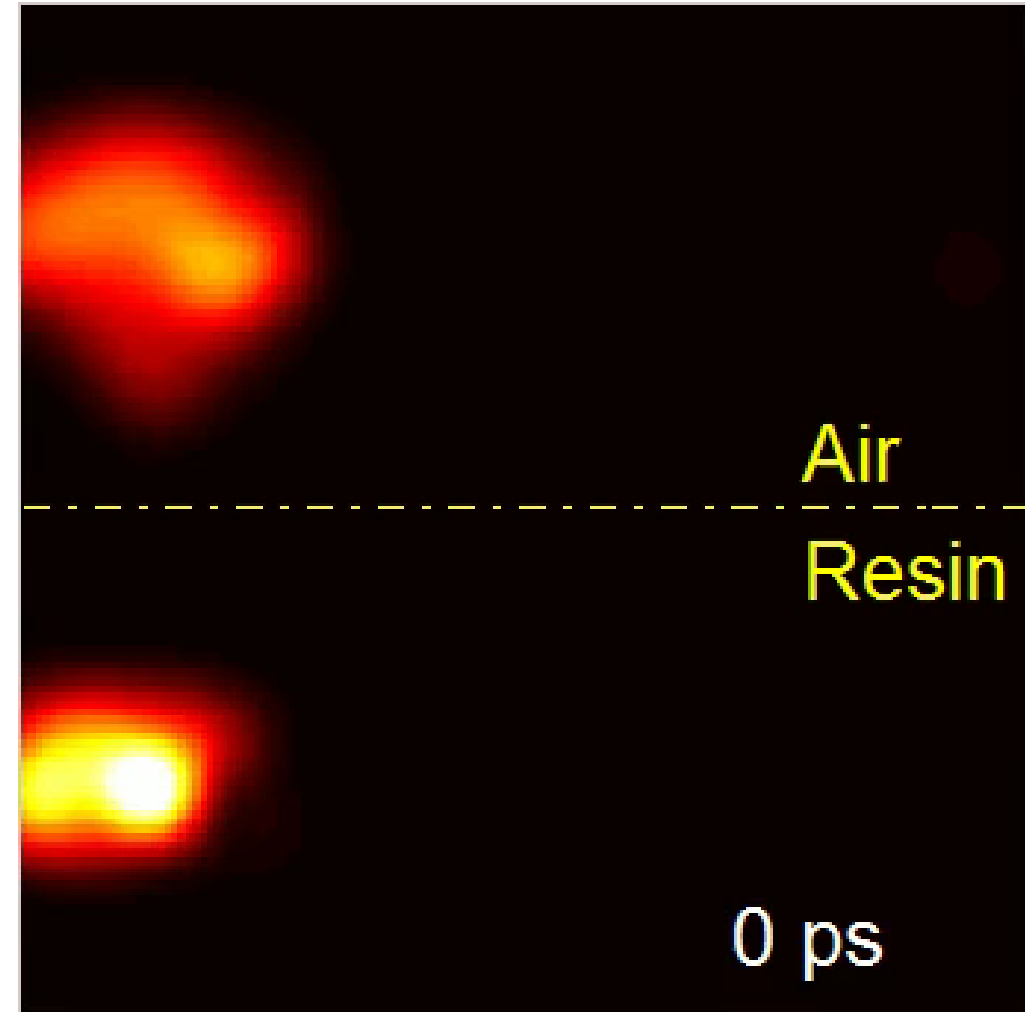
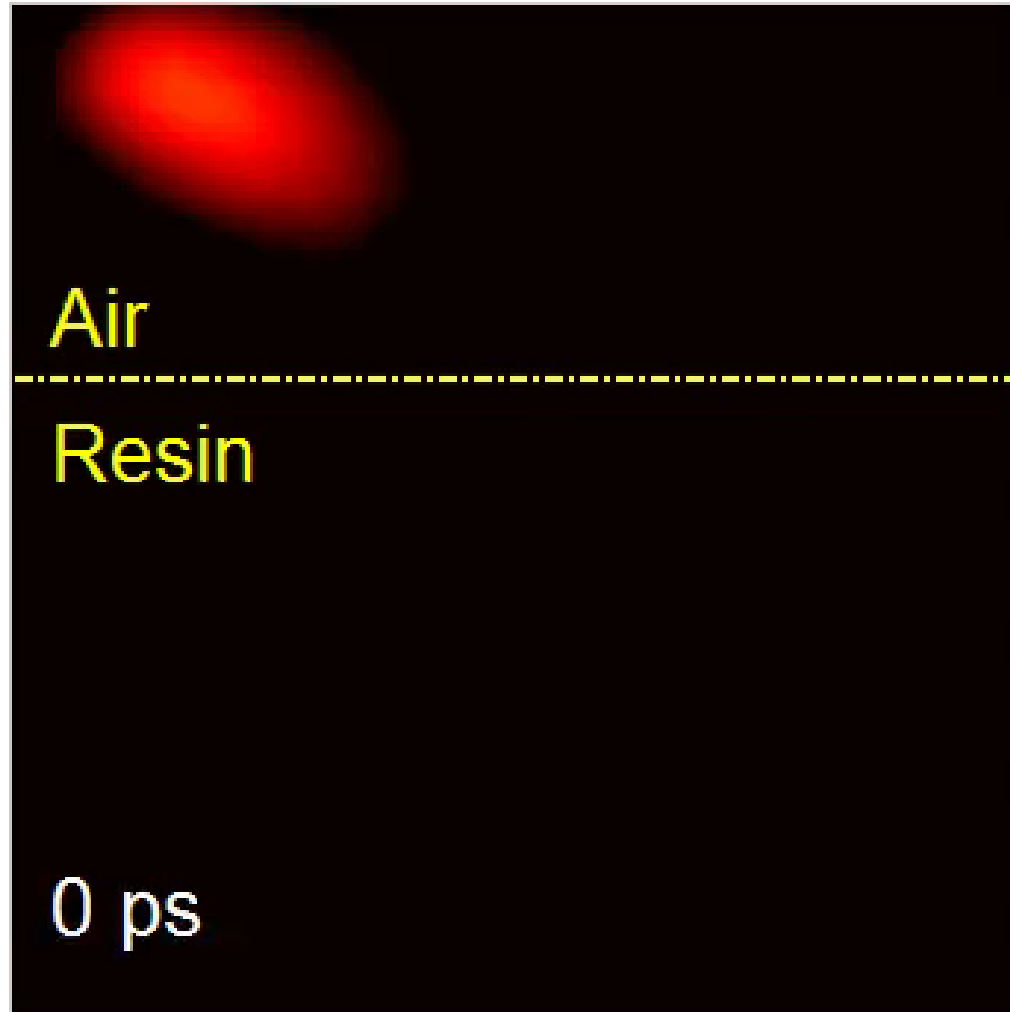
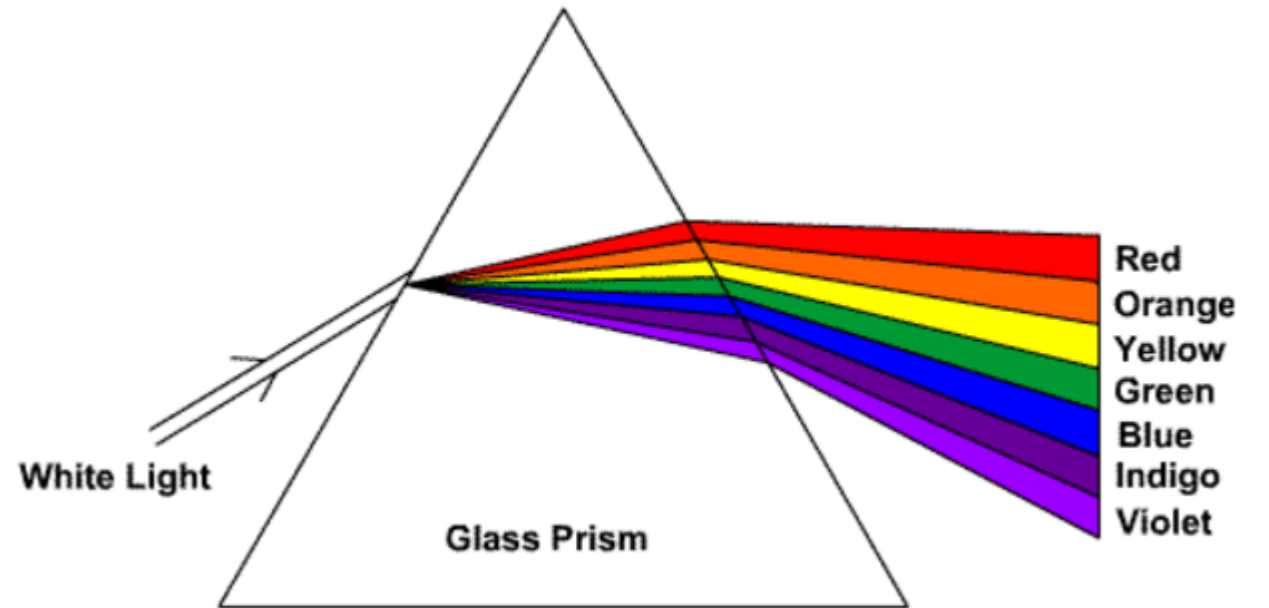


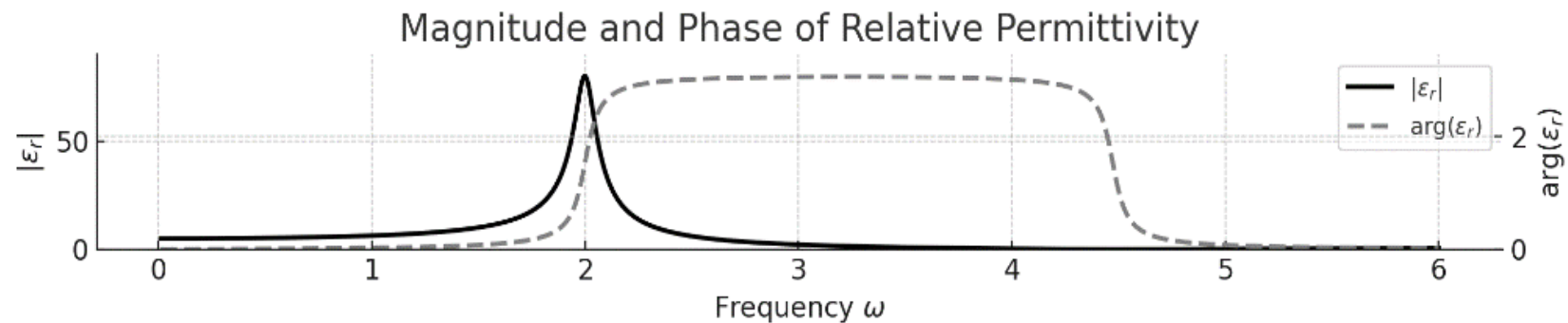
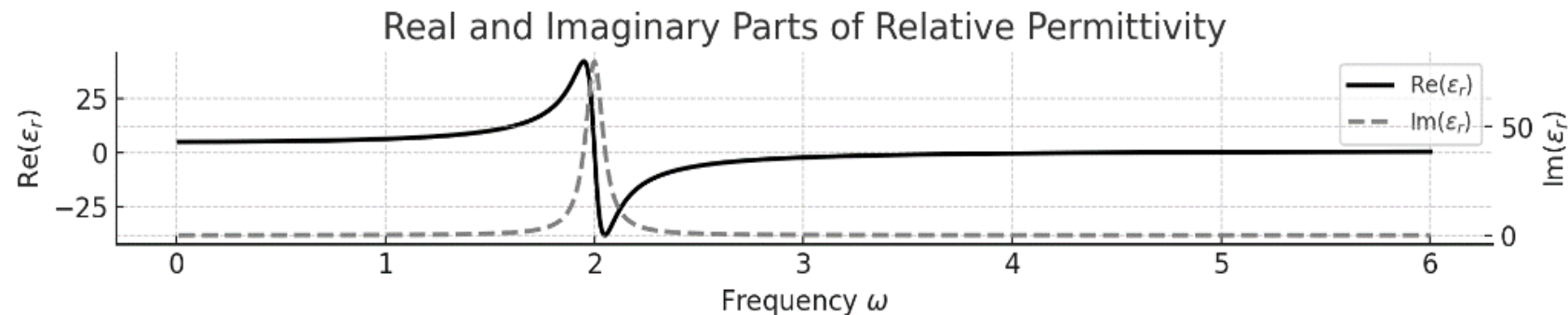
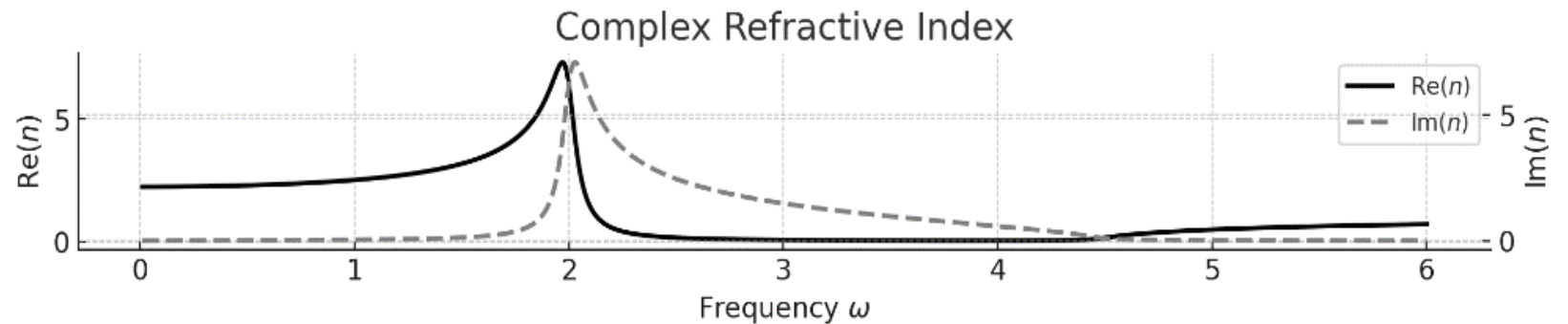
The existence of refractive index is weird



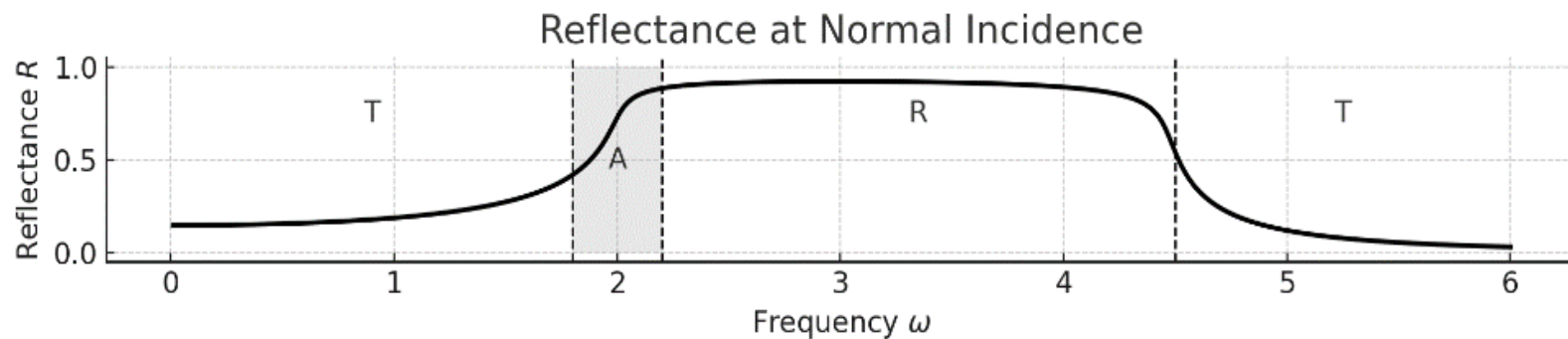
Light dispersion

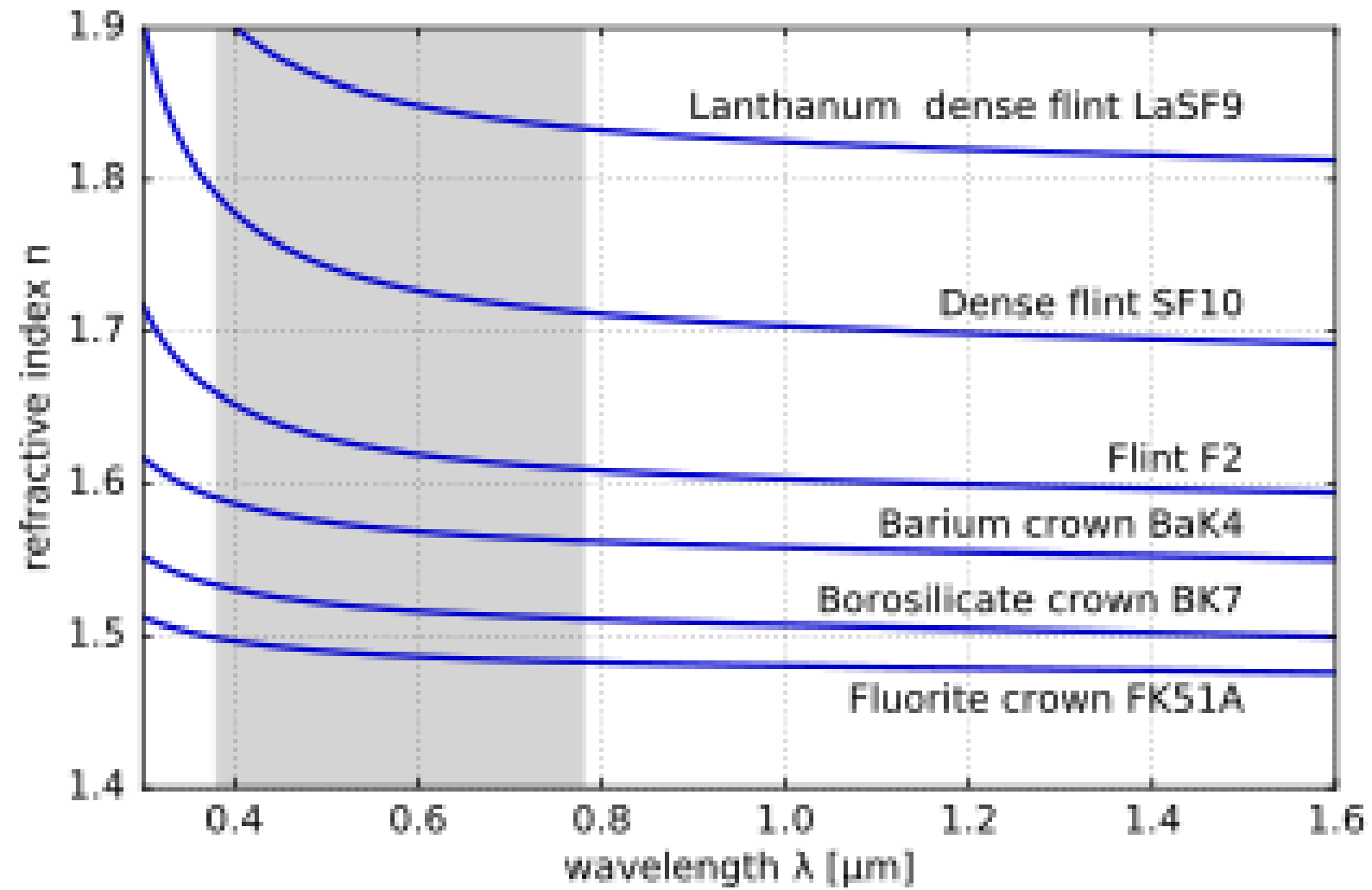


$$\omega_p = 4, \omega_0 = 2, \gamma = 0.1$$

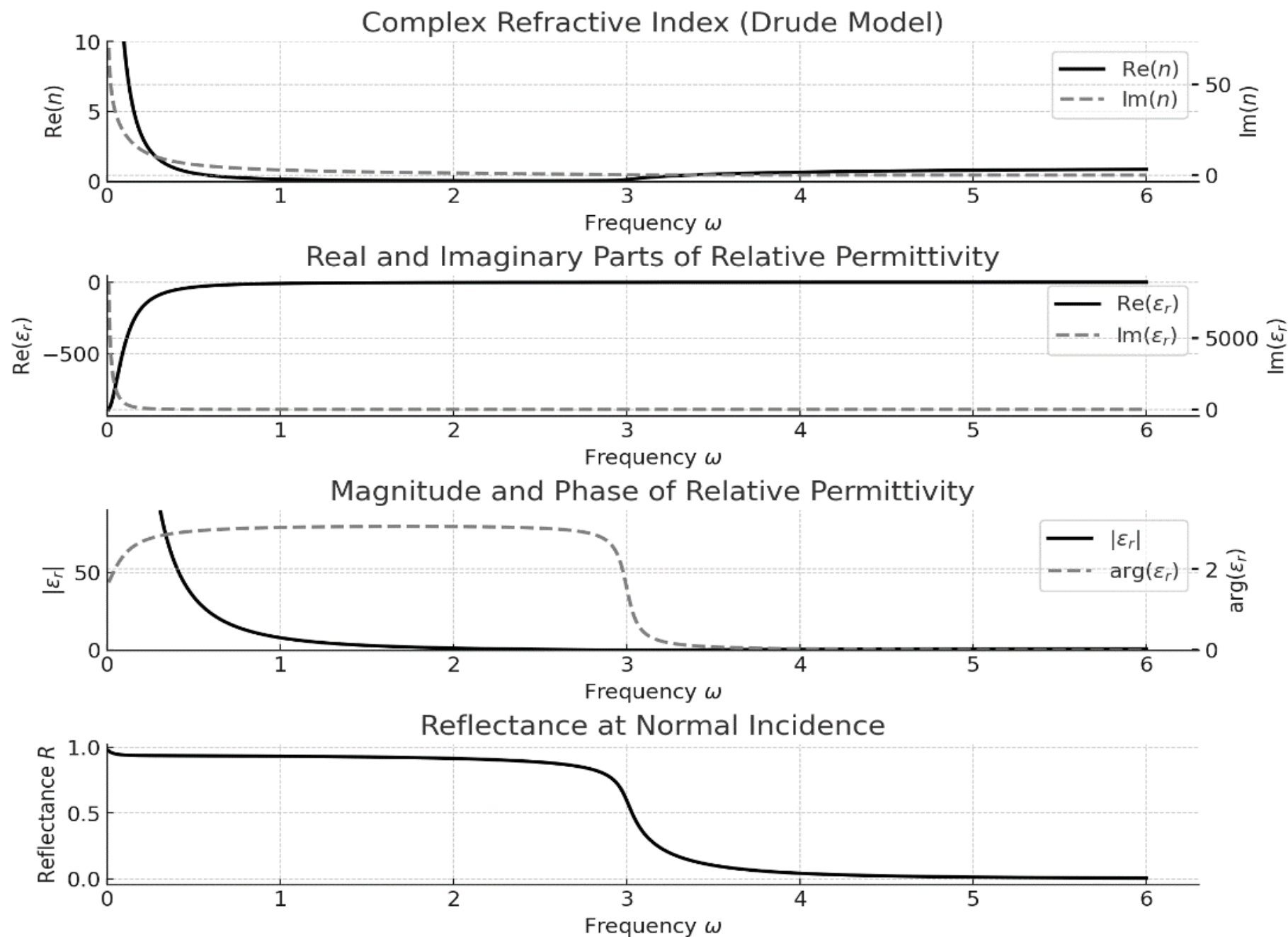


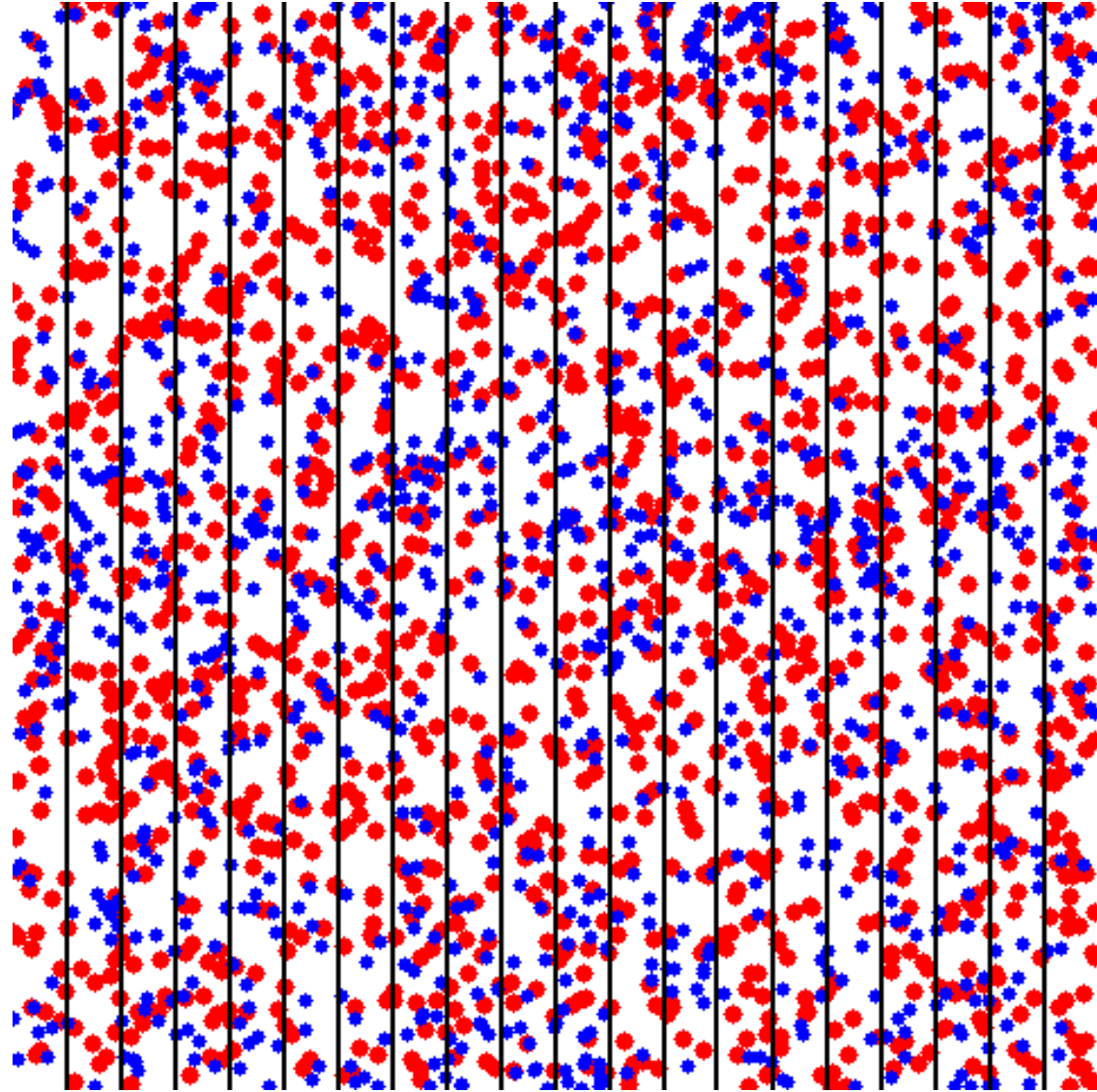
$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} \longrightarrow$$





$$\omega_p = 3, \omega_0 = 0, \gamma = 0.1$$





Natural plasma oscillation in metals

(<https://nextjournal.com/mpd/jupyter-pic-electron-plasma-wave-dispersion>)

For metals, the plasma frequency is a 'magic' turning point $\omega_p = \sqrt{\frac{Nq_e^2}{\epsilon_0 m}}$

For **copper**, $N = 8.5 \times 10^{28} m^{-3}$, $q_e = 1.6 \times 10^{-19} C$, $\epsilon_0 = 8.85 \times 10^{-12} F / m$

$$m = 9.11 \times 10^{-31} kg$$

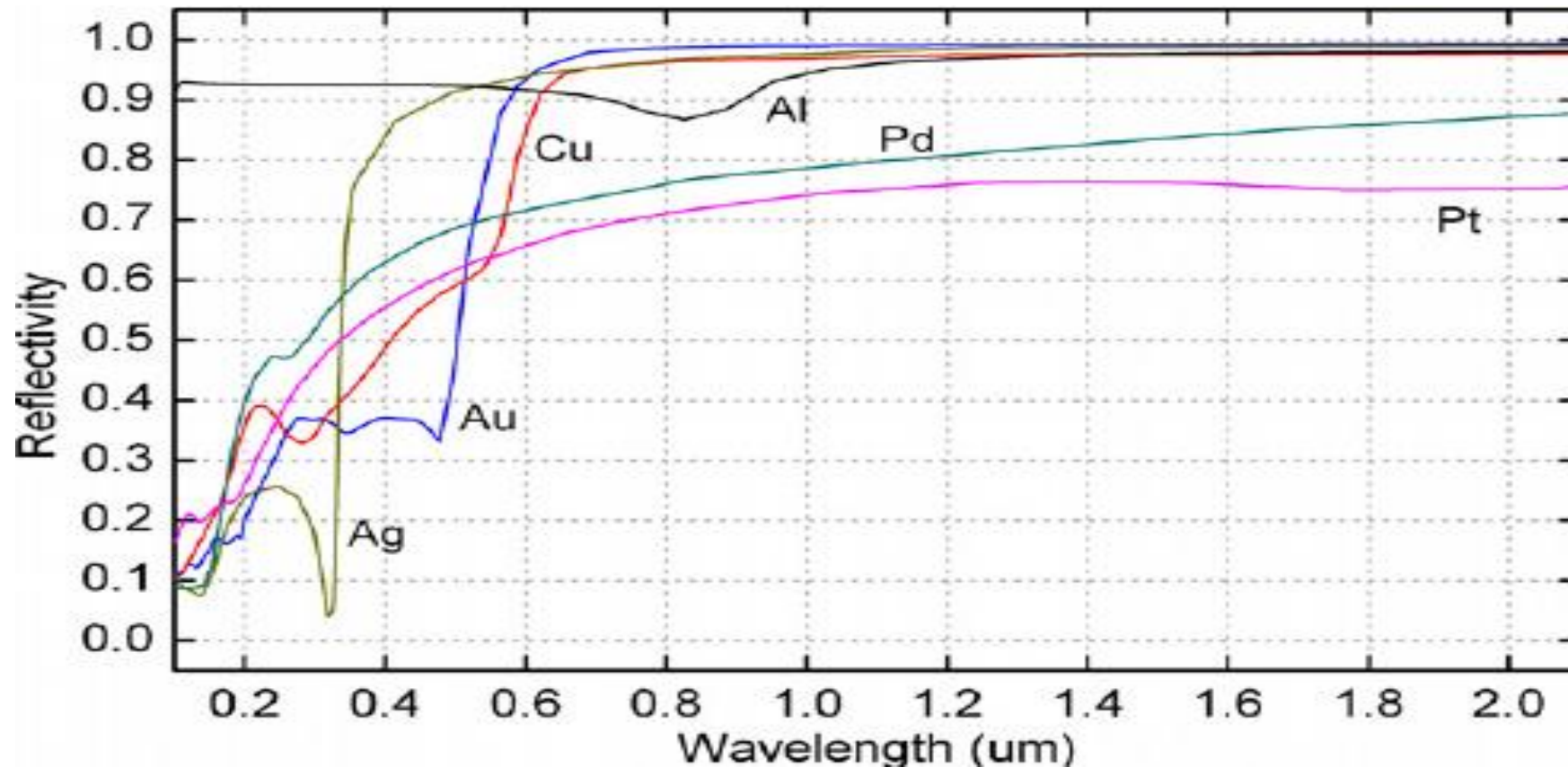
➔ $\omega_p = 16.4 \times 10^{15} s^{-1}$

Real value: $13.4 \times 10^{15} s^{-1}$, corresponding to **115 nm**

Case 1: For high frequency, ω on the same order of ω_p :

Example 1: Copper $\tau = 2.4 \times 10^{-14} \text{ s}$, $\gamma = \frac{1}{\tau} = 4 \times 10^{13} \text{ s}^{-1}$ $\omega_p = 16.4 \times 10^{15} \text{ s}^{-1}$

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega} \quad \longrightarrow \quad \boxed{\tilde{n}^2(\omega) \approx 1 - \frac{\omega_p^2}{\omega^2}}$$



Case 1: For high frequency, ω on the same order of ω_p :

Example 2: Ionosphere $\gamma = \frac{1}{\tau} = 10^5 \text{ s}^{-1}$ $\omega_p = 56 \times 10^6 \text{ s}^{-1}$ ($f_p = 9 \text{ MHz}$)

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}$$



$$\tilde{n}^2(\omega) \approx 1 - \frac{\omega_p^2}{\omega^2}$$

AM: 0.5 to 1.7 MHz

FM: 88 to 108 MHz



Case 2: For low frequency, $\omega \ll \omega_p$, $\omega \ll \gamma$ $\rightarrow \tilde{n}^2(\omega) \approx i \frac{\omega_p^2}{\gamma \omega}$

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega} \rightarrow \tilde{n}(\omega) \approx \sqrt{\frac{\omega_p^2}{\gamma\omega}} \cdot \frac{1}{\sqrt{2}} (1 + i) \rightarrow n \approx \kappa \approx \frac{\omega_p}{\sqrt{2\gamma\omega}}$$

Attenuation
coefficient:

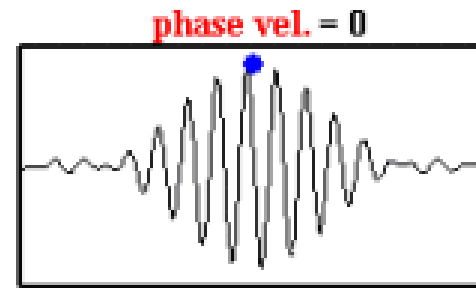
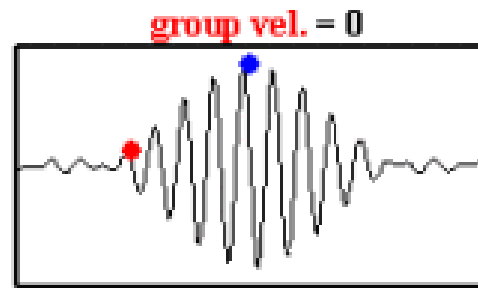
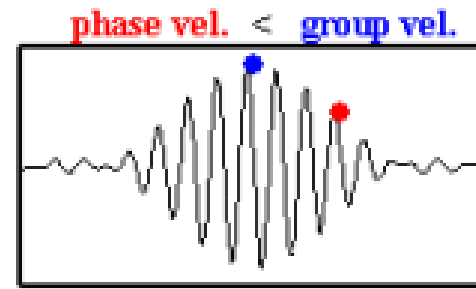
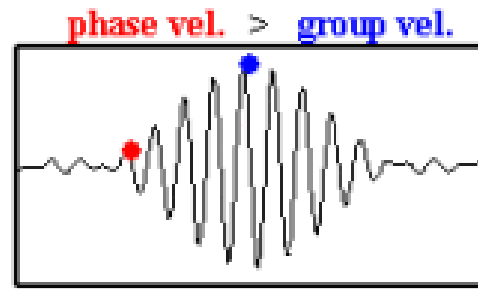
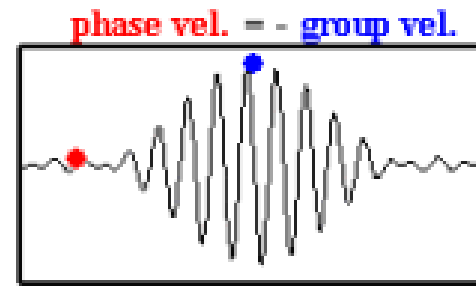
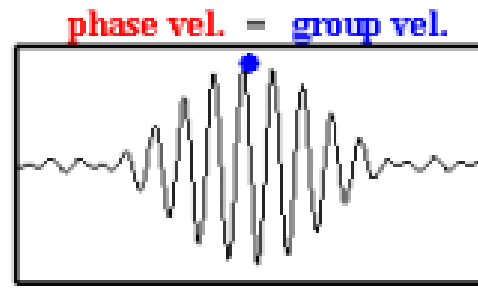
For **copper**,

$$N = 8.5 \times 10^{28} \text{ m}^{-3}, q_e = 1.6 \times 10^{-19} \text{ C}, \epsilon_0 = 8.85 \times 10^{-12} \text{ F / m} \quad m = 9.11 \times 10^{-31} \text{ kg} \quad \tau = 2.4 \times 10^{-14} \text{ s},$$

$$\rightarrow \omega_p = 16.4 \times 10^{15} \text{ s}^{-1} \quad \gamma = \frac{1}{\tau} = 4 \times 10^{13} \text{ s}^{-1}$$

For $f = 100 \text{ MHz}$ wave...

Phase velocity and group velocity



isvr